Alfredo Antonio Artiles Vallecillo

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Summary

Software Engineer with five years of experience in the mobility sector, specializing in C++ and Linux. Experience in international railway mobility and ticketing system projects in Ireland and Egypt, contributing to both software development and coordination and technical leadership activities.

Experience

Indra Group Madrid, Spain
C/C++ Software Engineer *Sept 2024 – Present*

- Assumed technical leadership and management responsibilities, coordinating tasks and incidents in international ticketing and validation projects in both Ireland and Egypt.
- Led the development team on the Irish ticketing project, including task assignment and tracking, technical coordination, and incident management.
- Developed software for the Egyptian railway project involving ticket vending machines and contactless card validation systems.
- Integrated peripheral devices through I2C protocols, serial communication, and IP-level network implementations, including banknote and coin validators, sensors, LEDs, 5G modems, and GPS/GNSS systems.
- Developed cross-compilable C++ software for multiple architectures using Jenkins and continuous integration processes.

Ceit Technology Center San Sebastian, Spain
Researcher *Sept 2021 – Sept 2024*

- Participated in Eurail, international European research projects with an investment exceeding 100 million euros, focused on the design, implementation, and standardization of communication protocols with leading companies in the railway sector.
- Developed and registered Python software for detecting train-carriage license plates in motion, adapted to images of varying quality and achieving up to 93% correct readings across more than 400 plates during a seven-day pilot.
- Developed and registered REST API software to emulate interference in railway communications based on real-world parameters, using `tc` and Linux kernel tools to configure delay, jitter, and packet loss.
- Developed a C++ remote train control system using MQTT and OPC-UA to interact with autonomous driving systems.
- Developed C++ train positioning systems, including PostgreSQL database access and communication with cloud services through MQTT.

Education

TECNUN, University of Navarra San Sebastian, Spain
B.Sc. in Industrial Electronic Engineering *Jun 2021*

- **Grade average:** 7.9 BOE
- **Highest Honors:** Computer Science II (C++), Electronic Circuits

Skills

Languages: C++, Python, Java, JavaScript
Technologies and tools: Linux, Docker, Git, MQTT, PostgreSQL, Jenkins, I2C, Serial, REST, OPC-UA
Spoken languages: Spanish (Native), English (TOEFL 111), French (B1)